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Question 1

Not answered

Marked out of 1

To be called a confounder, the third variable that is suspected to mediate the relationship between a cause and an effect must satisfy which of the following criteria?

Select one:

- ☐ It must precede the effect
- ☐ It must be related to both the cause and the effect
- ☐ It should be on the causal pathway that is being investigated
- ☐ It must be numerically quantifiable
- ☐ It must be consistently present along with the causal event

Your answer is incorrect.

A confounder is an extraneous third variable that correlates with both the dependent and independent variables in a study. Restriction, randomization, standardization, stratification and multivariate methods are various ways of eliminating the effect of a confounder.

The correct answer is: It must be related to both the cause and the effect

Question 2

Not answered

Marked out of 1

Which of the following is not a suitable method to reduce confounders in a study before the data is collected?

Select one:

- ☐ Restriction
- ☐ Randomisation
- ☐ Stratification
- ☐ Regression
- ☐ Minimisation

Your answer is incorrect.

Restriction, randomization, standardization, stratification and multivariate methods are various ways of eliminating the effect of a confounder. Of these multivariate methods such as regression can be used only after data collection, at the stage of analysis.

The correct answer is: Regression



Question 3

Not answered

Marked out of 1

Which of the following conditions of causality make cohort studies better than case-control studies in exploring causal association?

Select one:

- ☐ Consistency of association
- ☐ Strength of association
- ☐ Temporal association between cause and outcome
- ☐ Dose response relationship
- ☐ Biological plausibility of association

Your answer is incorrect.

The conditions of causality, as proposed by Bradford Hill, include 1. temporal association. 2. dose-response association 3. specificity of association 4. consistency of association 5. plausibility of biological reasoning 6. strength of association. Among this temporal association could not be demonstrated using a case-control design which is retrospective.

The correct answer is: Temporal association between cause and outcome



Question 4

Not answered

Marked out of 1

A statistician explains to you that confounders are not eliminated by randomisation. But he still insists randomisation is the best way of controlling confounding bias. Which of the following is a suitable explanation?

Select one:

- ☐ Statistically checking for confounders is not necessary if adequate randomisation is done
- ☐ Confounders can always be eliminated
- ☐ Confounders can cause an effect only if they are distributed differently between study groups
- ☐ Confounders are identified and restricted by the randomisation process
- ☐ Confounders can be statistically checked only in randomised samples

Your answer is incorrect.

Confounders can cause an effect only if they are distributed differently between study groups. This is the principle behind randomisation to reduce confounders.

The correct answer is: Confounders can cause an effect only if they are distributed differently between study groups



Question 5

Not answered

Marked out of 1

A study finds that the risk of developing hyperglycaemia is high in outpatients taking antipsychotics compared to antidepressants. The odds ratio is higher for a subgroup of psychotic patients with family history of diabetes. Family history of diabetes can be considered as

Select one:

- ☐ Confounder
- ☐ Systematic error
- ☐ Falsifier
- ☐ Effect modifier
- ☐ Causal factor

Your answer is incorrect.

Research studies tend to analyse the relationship between a cause and effect. A third factor can influence this relationship. This third factor is called a confounder if it distorts the relationship and produces an incorrect result. The confounding factor can disturb the association only if it is distributed unevenly in exposed and unexposed individuals. By contrast, effect modifier is a third factor that affects the magnitude of the cause-effect relationship. Effect modification provides important information and is not related to unequal distribution. One way of differentiating between confounders and effect modifiers is by studying crude relative risk and stratified relative risk. If crude risk differs from adjusted risk, we can conclude that the stratified factor is a confounder. If the relative risk of one stratum is different from other strata (subgroups) then we can call the factor as an effect modifier. Sometimes, a factor can be both confounder and effect modifier.

The correct answer is: Effect modifier

Question 6

Not answered

Marked out of 1

Which of the following statistical procedures involve stratified analysis of odds ratio in the presence of confounding?

Select one:

- ☐ Kruskal Wallis test
- ☐ ANOVA
- ☐ Mantel-Haenszel method
- ☐ Peto test
- ☐ Mann Whitney U test

Your answer is incorrect.

The Mantel-Haenszel technique is an excellent method for adjusting for confounding factors when analysing the relationship between a dichotomous outcome and a dichotomous risk factor.

The correct answer is: Mantel-Haenszel method



Question 7

Not answered

Marked out of 1

Which of the following methods is NOT used to minimise confounding factors in a case-control study?

Select one:

- ☐ Regression analysis
- ☐ Stratified analysis
- ☐ Blinding
- ☐ Matching
- ☐ Restriction

Your answer is incorrect.

Randomisation, restriction and matching are three methods that can be used while designing a study to control for confounders. Alternatively, we can handle the problem of confounding in the data analysis in one of the two ways: stratification or adjustment.

The correct answer is: Blinding



Question 8

Not answered

Marked out of 1

Various study designs can be used to investigate causality. In such investigations, the most important difference between an experimental study and an observational study is that

Select one:

- ☐ The researcher controls the exposure of the subjects in an experimental study
- ☐ The study and control groups are selected according to exposure status in an experimental study
- ☐ The experimental study needs ethical approval
- ☐ The experimental study is prospective
- ☐ The study and control groups are carefully matched in an experimental study

Your answer is incorrect.

In experimental studies, the exposure of the subjects to a putative causal factor / treatment factor is controlled by the researcher. Both types of studies need ethical approval. Cohort studies are observational and still prospective. Matching two groups is necessary for both types of studies. This can be achieved by randomization in experiments.

The correct answer is: The researcher controls the exposure of the subjects in an experimental study



Question 9

Not answered

Marked out of 1

While conducting a study, all enrolled subjects were matched for age, gender, smoking and body mass index before studying to control for confounding. Which of the following type of studies is most suitable for the above method?

Select one:

- ☐ A cross-sectional study design to study interaction between multiple social and environmental variables on adolescent depression
- ☐ An ecological study using data from the UK household survey
- ☐ A prospective cohort study of 1684 subjects.
- ☐ A case control study requiring a total sample size of 30 each.
- ☐ A randomised control trial evaluating the effect of Bupropion on smoking cessation.

Your answer is incorrect.

Matching is possible for smaller studies. It is not needed for RCTs and extremely difficult for large cohort studies. It is not possible for ecological studies and cross-sectional surveys.

The correct answer is: A case control study requiring a total sample size of 30 each.



Question 10

Not answered

Marked out of 1

Which of the following is not a criterion used in assessing the evidence for causality in medical research?

Select one:

- ☐ Temporality
- ☐ Sensitivity
- ☐ Consistency
- ☐ Specificity
- ☐ Biological gradient

Your answer is incorrect.

Hill's criteria for causality includes consistency, specificity, temporality and biological gradient (dose-response relationship). Consistency refers to the association being repeatedly observed in studies performed by different persons, in different settings, among different populations and using different methods. If a specific exposure can be isolated from others and associated with a specific disease, then such specificity supports causality. This is perhaps the most difficult criterion to fulfill because in practice many exposures (e.g., cigarettes or radiation) are associated with multiple effects and specific diseases often have more than one cause. Temporality refers to time relationship between cause and effect: the factor believed to have caused the disease must have occurred prior to disease development. Hill AB: The environment and disease: Association or causation? Proc R Soc Med 58:295-330, 1965

The correct answer is: Sensitivity

Question 11

Not answered

Marked out of 1

An observed dose-response effect supports the hypothesis of a causal relationship, according to Hill's criteria. But many well-designed studies of a genuinely causal relationship may fail to show a dose-response effect. Which is the least likely explanation for such results?

Select one:

- ☐ All doses are too low to reveal the effect
- ☐ A confounder obscures the effect
- ☐ Random variation obscures the effect
- ☐ There is no real dose-response effect
- ☐ All doses are too high to reveal the effect

Your answer is incorrect.

If a dose-response relationship is present, it provides a strong evidence for a causal relationship. However, the absence of a dose-response relationship does not necessarily rule out a causal relationship. In some cases in which a threshold may exist, no disease may develop up to a certain level of exposure; above this threshold level, the disease may manifest. The study might have included only below threshold exposures; similarly if there is a dose window only within which the response occurs, again this may be missed if all participants are outside the window. Similarly confounders and random variation may obscure the association.

The correct answer is: There is no real dose-response effect

Question 12

Not answered

Marked out of 1

Which of the following methods can be used to evaluate both confounders and effect modifiers in a study examining causal association

Select one:

- ☐ Standardisation
- ☐ Minimisation
- ☐ Randomisation
- ☐ Restriction
- ☐ Stratification

Your answer is incorrect.

Stratification can help analyse the effect modifiers and confounders in a study.

The correct answer is: Stratification



Question 13

Not answered

Marked out of 1

In a large Finnish cohort study of the relationship between bullying and depression in adult life, the crude relative risk was 5.0. When the data was stratified according to socioeconomic class of parents, the relative risks are 7.0 for the class I, 3.0 for the class II, and 1.5 for the class III and the Mantel-Haenszel adjusted rate was 3. The most appropriate conclusion from the study must be

Select one:

- ☐ Bullying is associated with three times more risk of depression in adult life; the risk is confounded by socioeconomic status of parents
- ☐ Bullying is associated with three times more risk of depression in adult life; the risk is modified according to socioeconomic status of parents
- ☐ Bullying is associated with 11.5 times more risk of depression in adult life
- ☐ Bullying is associated with four times (on average across various socioeconomic groups) more risk of depression in adult life
- ☐ Bullying is associated with 3.8 times more risk of depression in adult life

Your answer is incorrect.

As crude rates do not differ from adjusted rates, the socioeconomic status is not a confounder, but as strata differ among each other, it is an effect modifier. One must use the adjusted rates when giving the final study result - averages or sums are not appropriate.

The correct answer is: Bullying is associated with three times more risk of depression in adult life; the risk is modified according to socioeconomic status of parents

Question 14

Not answered

Marked out of 1

A researcher is interested in the effects of alcohol use on REM sleep architecture. But she is aware that cannabis or nicotine use is a potential extraneous variable; so she picks only those subjects who do not use cannabis or nicotine before determining their alcohol status. Which of the following methods of controlling a confounder has been used here?

Select one:

- ☐ Restriction
- ☐ Stratification
- ☐ Standardisation
- ☐ Randomisation
- ☐ Matching

Your answer is incorrect.

The study has been restricted to a population group lacking the confounding factor, namely exposure to cannabis or nicotine. This is known as restriction.

The correct answer is: Restriction



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